Exam

Monday, March 16, 2026 9:22 PM

1. $|\psi\rangle = \sqrt{\frac{1}{5}}|0\rangle - \sqrt{\frac{2}{5}}|1\rangle + \sqrt{\frac{1}{5}}|2\rangle + \sqrt{\frac{1}{5}}|3\rangle$

a. $\mathbb{P}(\text{observe } |1\rangle) = |\langle 1|\psi\rangle|^2 = \left|-\sqrt{\frac{2}{5}}\right|^2 = \boxed{\frac{2}{5}}$

b. $|v_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ $\qquad |v_3\rangle = \frac{1}{\sqrt{2}}(|2\rangle - |3\rangle)$

$\mathbb{P}(\text{observe } |v_1\rangle) = |\langle v_1|\psi\rangle|^2$

$\langle v_1|\psi\rangle = \frac{1}{\sqrt{2}}(\langle 0| - \langle 1|)\left(\sqrt{\frac{1}{5}}|0\rangle - \sqrt{\frac{2}{5}}|1\rangle + \sqrt{\frac{1}{5}}|2\rangle + \sqrt{\frac{1}{5}}|3\rangle\right)$

$= \frac{1}{\sqrt{2}}\left(\sqrt{\frac{1}{5}} + \sqrt{\frac{2}{5}}\right) = \frac{1}{\sqrt{2}} \frac{1+\sqrt{2}}{\sqrt{5}} = \frac{1+\sqrt{2}}{\sqrt{10}}$

$\mathbb{P}(\text{observe } |v_1\rangle) = \left(\frac{1+\sqrt{2}}{\sqrt{10}}\right)^2 = \frac{1}{10}(1 + 2\sqrt{2} + 2) = \boxed{\frac{3+2\sqrt{2}}{10}}$

$\mathbb{P}(\text{observe } |v_3\rangle) = |\langle v_3|\psi\rangle|^2$

$\langle v_3|\psi\rangle = \frac{1}{\sqrt{2}}(\langle 2| - \langle 3|)\left(\sqrt{\frac{1}{5}}|0\rangle - \sqrt{\frac{2}{5}}|1\rangle + \sqrt{\frac{1}{5}}|2\rangle + \sqrt{\frac{1}{5}}|3\rangle\right)$

$= \frac{1}{\sqrt{2}}\left(\sqrt{\frac{1}{5}} - \sqrt{\frac{1}{5}}\right) = 0$

$\mathbb{P}(\text{observe } |v_3\rangle) = \boxed{0}$

2a. $|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle - |10\rangle - |11\rangle) = \frac{1}{2}(|0\rangle(|0\rangle + |1\rangle) - |1\rangle(|0\rangle + |1\rangle))$

$= \frac{1}{2}(|0\rangle - |1\rangle)(|0\rangle + |1\rangle) = \underbrace{\boxed{\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)}}_{|q_1\rangle} \otimes \underbrace{\boxed{\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)}}_{|q_2\rangle}$

b. $|\psi\rangle = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle - |11\rangle)$

assume for the sake of contradiction that $|\psi\rangle$ is not entangled, ie $|\psi\rangle = |q_1\rangle \otimes |q_2\rangle$ where $|q_1\rangle = \alpha|0\rangle + \beta|1\rangle$ and $|q_2\rangle = \gamma|0\rangle + \delta|1\rangle$.

then $|\psi\rangle = (\alpha|0\rangle + \beta|1\rangle)(\gamma|0\rangle + \delta|1\rangle) =$

$= \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle$

then $\alpha\gamma = \frac{1}{2}$ $\quad \alpha\delta = -\frac{1}{2}$ $\quad \beta\gamma = -\frac{1}{2}$ $\quad \beta\delta = -\frac{1}{2}$

then $\beta\gamma = \beta\delta$, so $\gamma = \delta$.

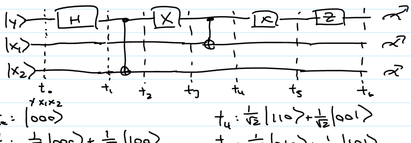
but also $\alpha\gamma = -\alpha\delta$, so $\gamma = -\delta$. then we must have $\gamma = \delta = 0$.

but for we have a contradiction because $\alpha\gamma = \frac{1}{2}$ and $\alpha\gamma = 0$.

so $|\psi\rangle$ must be entangled. □

3. $x = \{0,1\}^2$ $\qquad y = \{0,1\}$ $\qquad |g(x,y)\rangle = \frac{1}{\sqrt{2}}|0,x\rangle + (-1)^y \frac{1}{\sqrt{2}}|1,\bar{x}\rangle$

build circuit st $|0,0,0\rangle \xrightarrow[\text{circuit}]{} |g(1,1)\rangle = \frac{1}{\sqrt{2}}|0,1,0\rangle - \frac{1}{\sqrt{2}}|1,0,1\rangle$



t_0: $|000\rangle$

t_1: $\frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|100\rangle$

t_2: $\frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|101\rangle$

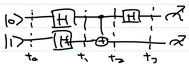
t_3: $\frac{1}{\sqrt{2}}|100\rangle + \frac{1}{\sqrt{2}}|001\rangle$

t_4: $\frac{1}{\sqrt{2}}|110\rangle + \frac{1}{\sqrt{2}}|001\rangle$

t_5: $\frac{1}{\sqrt{2}}|010\rangle + \frac{1}{\sqrt{2}}|101\rangle$

t_6: $\frac{1}{\sqrt{2}}|010\rangle - \frac{1}{\sqrt{2}}|101\rangle$

4.



t_0: $|01\rangle$

t_1: $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$

t_2: $\frac{1}{2}(|00\rangle - |01\rangle + |11\rangle - |10\rangle) = \frac{1}{2}(|0\rangle(|0\rangle - |1\rangle) + |1\rangle(|1\rangle - |0\rangle))$

$= \frac{1}{2}(|0\rangle - |1\rangle)(|0\rangle - |1\rangle)$

t_3: $\frac{1}{\sqrt{2}}|1\rangle(|0\rangle - |1\rangle) = \boxed{\frac{1}{\sqrt{2}}(|10\rangle - |11\rangle)}$

$\boxed{\mathbb{P}(\text{measure } |00\rangle) = 0}$

5a. $f: \{0,1\}^n \to \{0,1\}^m$ $\qquad U_f|\overset{n}{x}, \overset{m}{y}\rangle = |x, f(x) \oplus y\rangle$

given $|z,w\rangle$, $z \in \{0,1\}^n$ $w \in \{0,1\}^m$, find x,y

st $U_f|x,y\rangle = |z,w\rangle$ (show U_f invertible)

$|x, f(x) \oplus y\rangle = |z,w\rangle$ $\qquad \boxed{x = z}$

$f(z) \oplus y = w$ $\qquad \boxed{y = w \oplus f(z)}$

b. $|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x, 0^m\rangle$

$U_f|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} U_f|x, 0^m\rangle = \boxed{\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x, f(x)\rangle}$

c. $I_y = \{x : f(x) = y\}$

measure output register (last m bits).

show $\mathbb{P}(\text{observe particular } y \in \{0,1\}^m) = \frac{1}{|I_y|}$. write post-measured state.

$|\psi\rangle = \sum_y |e_y, y\rangle$ $\qquad \mathbb{P}(\text{observe particular } y) = \langle e_y, e_y\rangle$

post-measurement state: $\frac{|e_y, y\rangle}{\sqrt{P(y)}}$

$U_f|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x, f(x)\rangle$

We want to find the probability of observing some y,

So we rewrite the sum in terms of y.

$U_f|\psi\rangle = \sum_{y \in \{0,1\}^m} \left(\sum_{x \in I_y} \frac{1}{\sqrt{2^n}}|x,y\rangle\right)$

y since $f(x) = y$

all x st $f(x) = y$

Then by the partial measurement rule,

$\mathbb{P}(y) = \left(\sum_{x \in I_y} \frac{1}{\sqrt{2^n}}\langle x|\right)\left(\sum_{x \in I_y} \frac{1}{\sqrt{2^n}}|x\rangle\right) = \sum_{x \in I_y} \frac{1}{2^n} = \boxed{\frac{|I_y|}{2^n}}$

since when $x = x'$, $\langle x|x'\rangle = 1$

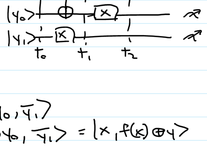
and otherwise $\langle x|x'\rangle = 0$

Post-measured-state: $\frac{1}{\sqrt{2^n}} \frac{\sum_{x \in I_y} |x,y\rangle}{\sqrt{\frac{|I_y|}{2^n}}} = \boxed{\frac{\sum_{x \in I_y} |x,y\rangle}{\sqrt{|I_y|}}}$

d.

x	$f(x)$
00	11
01	11
10	01
11	01

$U_f|x,y\rangle = |x, f(x) \oplus y\rangle$



t_0: $|x_0, x_1, y_0, y_1\rangle$

t_1: $|x_0, x_1, x_0 \oplus y_0, \overline{y_1}\rangle$

t_2: $|x_0, x_1, x_0 \oplus y_0, \overline{y_1}\rangle = |x, f(x) \oplus y\rangle$